

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1590

Tenaya 138 kV
October 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1590. The Interconnection Customer proposes a 101.5 MW battery energy storage facility interconnecting at the Tenaya 138 kV substation in the West Virginia area. The request was received on October 08, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 583 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,414,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bartletts Ferry – Santo Domingo 138 kV	Salisbury – Tenaya 138 kV	136.4	218	100.0
Dryburg – Santo Domingo 138 kV	Salisbury – Tenaya 138 kV	135.1	223	100.0
Dryburg – Tenaya 138 kV line	Salisbury – Tenaya 138 kV	134.5	226	100.0
Tenaya – Salisbury Substation 138 kV Line	Dryburg – Tenaya 138 kV	133.0	228	100.0
Spencer Switch – Salisbury 138 kV	Dryburg – Tenaya 138 kV	130.5	230	100.0
Spencer Switch Substation – Central Ephrata 138 kV line	Dryburg – Tenaya 138 kV	128.1	232	100.0
7ThAvenue – Northwest Beloit 230 kV	Valemount – Tristate 345 kV	100.2	548	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bartletts Ferry – Santo Domingo 138 kV	Rebuild line with higher-capacity conductor	\$2,955,000	33% (\$983,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dryburg – Santo Domingo 138 kV	Rebuild line with higher-capacity conductor	\$5,554,000	33% (\$1,847,000)
Dryburg Substation – Tenaya 138 kV	Rebuild line with higher-capacity conductor	\$9,876,000	33% (\$3,284,000)
Salisbury – Tenaya 138 kV L	Rebuild line with higher-capacity conductor	\$6,456,000	33% (\$2,147,000)
Spencer Switch – Salisbury 138 kV	Rebuild line with higher-capacity conductor	\$8,203,000	33% (\$2,728,000)
Spencer Switch – Central Ephrata 138 kV Line	Rebuild line with higher-capacity conductor	\$7,133,000	33% (\$2,372,000)
7ThAvenue – Northwest Beloit 230 kV	Reconductor line	\$2,163,000	2% (\$53,000)
	Total allocated to project		\$13,414,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.